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CHARACTERIZATION OF WIRELESS NETWORK SLICE USAGE

Abstract

A data communication system characterizes the use of a wireless network slice by wireless communication devices. The data communication system receives requests from the wireless communication devices to use the wireless network slice. The data communication system determines that the use of the wireless network slice by the wireless communication devices would be unauthorized, and in response, serves the wireless communication devices using the wireless network slice. The data communication system identifies a characteristic of the wireless network slice responsive to serving the wireless communication devices using the wireless network slice.

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Background/Summary

TECHNICAL BACKGROUND

[0001] Wireless communication networks provide wireless data services to wireless

communication devices like phones, computers, and other user devices. The wireless data services may include internet-access, data messaging, video conferencing, or some other data communication product. The wireless communication networks comprise wireless access nodes like Wireless Fidelity (WIFI) hotspots, Fifth Generation New Radio (5G NR) cell towers, and satellites in earth orbit. The wireless communication networks further comprise network elements that process network signaling and handle user data like Access and Mobility Management Functions, Session Management Functions (SMFs), User Plane Functions (UPFs), and Unified Data Management (UDMs).

[0002] Wireless network slices comprise network elements like UPFs and SMFs that are customized for specific user applications and network services. The wireless communication devices request the specific wireless network slice for their current user application and network service. The wireless communication networks then deliver this network service using the requested wireless network slice. For example, a wireless communication device may use a low-latency wireless network slice to serve a low-latency user application like an augmented-reality display.

[0003] There are scenarios when the use of a wireless network slice by a wireless communication device would be unauthorized. The wireless communication device may not be subscribed to the wireless network slice. The wireless communication device may be subscribed to the wireless network slice, but for some reason the wireless communication network instructs the wireless communication device not to use the slice. A request to use the wireless network slice by the wireless communication devices after receiving such an instruction would be unauthorized. The wireless communication network may instruct the wireless communication device not to request a wireless network slice when it is requesting or using another wireless network slice. Thus, the simultaneous use of both wireless network slices is not allowed. In this case, a request by the wireless communication device to use one slice when it is requesting or using the other slice would be unauthorized.

[0004] Unfortunately, the wireless communication networks fail to efficiently identify characteristics of unauthorized slice usage. Moreover, the wireless communication networks are unable to effectively identify patterns of the unauthorized slice usage across multiple wireless communication devices. These failures harm network customers by consuming network resources that could be used by legitimate users. The failures also harm the wireless communication networks through a loss of reputation and trust by their customers.

TECHNICAL OVERVIEW

[0005] In some examples, the use of a wireless network slice by wireless communication devices is characterized. Requests from the wireless communication devices to use the wireless network slice are received. The use of the wireless network slice by the wireless communication devices is determined to be unauthorized. In response, the wireless communication devices are served using the wireless network slice. A characteristic of the wireless network slice is identified responsive to serving the wireless communication devices using the wireless network slice.

[0006] In some examples, a data communication system characterizes the use of a wireless network slice by wireless communication devices. The data communication system wirelessly broadcasts slice information related to the wireless network slice and receives requests from the wireless communication devices to use the wireless network slice. The wireless communication devices request the wireless network slice responsive to the wireless broadcast. The data communication system determines that the use of the wireless network slice by the wireless communication devices would be unauthorized, and in response, serves the wireless communication devices using the wireless network slice. The data communication system identifies a characteristic of the wireless network slice responsive to serving the wireless communication devices using the wireless network slice. The characteristic of the wireless network slice may comprise: types of the wireless communication devices, user applications in the wireless communication devices, operating

systems in the wireless communication devices, destination network addresses used by the wireless communication devices, uniform resource indicators used by the wireless communication devices, and geographic locations of the wireless communication devices.

[0007] In some examples, a data communication system characterizes the use of a wireless network slice by wireless communication devices. In the data communication system, control circuitry determines that the use of the wireless network slice by the wireless communication devices would be unauthorized, and in response, directs communication circuitry to serve the wireless communication devices using the wireless network slice. The communication circuitry serves the wireless communication devices using the wireless network slice responsive to the control circuitry. The control circuitry identifies a characteristic of the wireless network slice based on the wireless communication devices use of the wireless network slice.

Description

DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 illustrates an exemplary data communication system to characterize use of a wireless network slice.

[0009] FIG. 2 illustrates an exemplary operation of the data communication system to characterize the use of the wireless network slice.

[0010] FIG. 3 illustrates an exemplary operation of the data communication system to characterize the use of the wireless network slice.

[0011] FIG. 4 illustrates exemplary processing circuitry to characterize use of a wireless network slice.

[0012] FIG. 5 illustrates an exemplary wireless communication network to characterize unauthorized use of a wireless network slice.

[0013] FIG. 6 illustrates an exemplary UE in the wireless communication network that characterizes the unauthorized use of the wireless network slice.

[0014] FIG. 7 illustrates an exemplary Fifth Generation New Radio (5G NR) access node in the wireless communication network that characterizes the unauthorized use of the wireless network slice.

[0015] FIG. 8 illustrates an exemplary Wireless Fidelity (WIFI) access node in the wireless communication network that characterizes the unauthorized use of the wireless network slice.

[0016] FIG. 9 illustrates an exemplary satellite access node and ground station in the wireless communication network that characterizes the unauthorized use of the wireless network slice.

[0017] FIG. 10 illustrates an exemplary Network Function Virtualization Infrastructure (NFVI) in the wireless communication network that characterizes the unauthorized use of the wireless network slice.

[0018] FIG. 11 illustrates an exemplary operation of the wireless communication network to characterize the unauthorized use of the wireless network slice.

[0019] FIG. 12 illustrates an exemplary operation of the wireless communication network to characterize the unauthorized use of the wireless network slice.

[0020] FIG. 13 illustrates an exemplary operation of the wireless communication network to characterize the unauthorized use of the wireless network slice.

DETAILED DESCRIPTION

[0021] FIG. 1 illustrates exemplary data communication system **100** to characterize the use of wireless network slice **106**. Data communication system **100** comprises wireless communication devices **101-103**, control circuitry **104**, and communication circuitry **105**. Communication circuitry **105** comprises wireless network slice **106**.

[0022] In some examples, wireless communication device **101** transfers a request for wireless

network slice **106** to control circuitry **104** over communication circuitry **105**. Control circuitry **104** determines that the use of the wireless network slice **106** by wireless communication device **101** would be unauthorized. For example, wireless communication device **101** may not have a subscription to wireless network slice **106**. In response to the request and lack of authorization, control circuitry **104** directs communication circuitry **105** to serve wireless communication device **101** using wireless network slice **105**. Communication circuitry **105** serves wireless communication device **101** using wireless network slice **106** responsive to the direction of control circuitry **104**. Control circuitry **104** identifies a characteristic of wireless network slice **106** based on the use of wireless network slice **106** by wireless communication device **101**. For example, control circuitry **104** may identify a user application used by wireless communication device **101** when requesting wireless network slice **106**. In some examples, wireless network slice **106** only monitors unauthorized slice usage and does not have any authorized wireless communication devices. Data communication system **100** determines additional slice characteristics or patterns of slice characteristics by serving wireless communication devices **102-103** in the manner of wireless communication device **101**.

[0023] In some examples, communication circuitry **105** wirelessly broadcasts slice information related to the wireless network slice **106**, and wireless communication devices **101-103** request wireless network slice **106** responsive to the wireless broadcast. The wireless broadcast may indicate that wireless network slice **106** delivers a communication service that drives wireless communication devices **101-103** to request wireless network slice **106**. The communication service may comprise a high-throughput service, low-latency service, high-definition service, gaming service, telepresence service, or some other data network product. The identified characteristic of wireless network slice **106** may comprise: types of wireless communication devices **101-103**, user applications in wireless communication devices **101-103**, operating systems in wireless communication devices **101-103**, destination network addresses used by wireless communication devices **101-103**, uniform resource indicators used by wireless communication devices **101-103**, geographic locations of wireless communication devices **101-103**, or some other information related to the use of wireless network slice **106**.

[0024] A high-throughput service transfers user data at fast speeds for applications like software distribution. A low-latency service transfers user data with minimal delay between data blocks for applications like augmented reality. A high-definition service transfers large amounts of data per portion of media for applications like sports content streaming. A gaming service delivers interactive entertainment for applications like combat flight simulators. A telepresence service comprises video/audio distribution for applications like media conferencing. Types of wireless communication devices comprises model numbers, brand names, and the like. User applications comprise software modules that are executed in wireless user devices **101-103** and that consume slice services. Operating systems comprise software modules in wireless communication devices **101-103** that interface between user applications and device hardware. Destination network addresses are used to transfer data packets to communication end-points like web sites. Uniform resource indicators indicate end-points like web sites for data communication with wireless communication devices **101-103**. Geographic locations of wireless communication devices **101-103** comprise geo-coordinates, network sectors, tracking areas, or the like.

[0025] There are various scenarios when the use of wireless network slice **106** would be unauthorized. In some scenarios, the purpose of wireless network slice **106** is to characterize unauthorized use, so slice **106** does not have any authorized devices, and any request by wireless communication devices **101-103** to use wireless network slice **106** would be unauthorized. In other scenarios, wireless communication devices **101-102** are subscribed to wireless network slice **106**, but wireless communication device **103** is not subscribed to slice **106**, so requests to use slice **106** by wireless communication device **103** would be unauthorized. In yet other scenarios, wireless communication devices **101-103** may be subscribed to wireless network slice **106**, but for some

reason data communication system **100** instructs the wireless communication devices **101-103** not to use slice **106**, so requests to use wireless network slice **106** by wireless communication devices **101-103** after receiving such an instruction would be unauthorized. For example, wireless network slice **106** may not be available in the geographic area occupied by wireless communication devices **101-103**. Wireless network slice **106** may not be available due to maintenance, network outage, or some other reason, and requests to use slice **106** after an instruction not to would be unauthorized. In yet another example, data communication system **100** may instruct wireless communication devices **101-103** not to request wireless network slice **106** when they are requesting or using another wireless network slice. Thus, the simultaneous use of both wireless network slices is not allowed, so a request by wireless communication devices **101-103** to use slice **106** when they are requesting or using the other slice would be unauthorized.

[0026] Wireless communication devices **101-103** comprise communication equipment that is operated by a user like phones, computers, and/or some other data communication apparatus. Control circuitry **104** comprises one or more network elements that control the delivery of wireless communication services to wireless communication devices **101-103**. Control circuitry **104** may comprise an Access and Mobility Management Function (AMF), Unified Data Management (UDM), Application Server (AS), and/or some other network elements. Communication circuitry **105** comprises one or more network elements that deliver a communication service using wireless network slice **106**. Communication circuitry **105** may comprise wireless access nodes, Interworking Functions (IWFs), wireless network slices, and/or some other network elements. Wireless network slice **106** comprises one or more network elements that deliver a communication service to wireless communication devices **101-103**. Wireless network slice **106** comprises a Session Management Function (SMF), User Plane Function (UPF), and/or some other network elements.

[0027] Wireless communication devices **101-103** and communication circuitry **105** comprise radios that wirelessly communicate using wireless protocols like Wireless Fidelity (WIFI), Fifth Generation New Radio (5G NR), Long Term Evolution (LTE), Low-Power Wide Area Network (LP-WAN), Near-Field Communications (NFC), Code Division Multiple Access (CDMA), Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), and satellite data communications. Wireless communication devices **101-103** and circuitry **104-105** comprise microprocessors, software, memories, transceivers, bus circuitry, and/or some other data processing components. The microprocessors comprise Digital Signal Processors (DSP), Central Processing Units (CPU), Graphical Processing Units (GPU), Application-Specific Integrated Circuits (ASIC), and/or some other data processing hardware. The memories comprise Random Access Memory (RAM), flash circuitry, disk drives, and/or some other type of data storage. The memories store software like operating systems, utilities, protocols, applications, and functions. The microprocessors retrieve the software from the memories and execute the software to drive the operation of data communication system **100** as described herein.

[0028] FIG. 2 illustrates an exemplary operation of data communication system **100** to characterize the use of wireless network slice **106**. The operation may vary in other examples. Data communication system **100** receives requests from wireless communication devices **101-103** to use wireless network slice **106** (201). Data communication system **100** determines that the use of wireless network slice **106** by the wireless communication devices **101-103** would be unauthorized (202). In response to the determination, data communication system **100** serves wireless communication devices **101-103** using wireless network slice **106** (203). Data communication system **100** identifies a characteristic of wireless network slice **106** responsive to serving wireless communication devices **101-103** using wireless network slice **106** (204).

[0029] FIG. 3 illustrates an exemplary operation of data communication system **100** to characterize the unauthorized use of wireless network slice **106**. The operation may vary in other examples. Communication circuitry **105** wirelessly broadcasts slice information related to wireless network

slice **106**. Wireless communication device **101** receives the wireless broadcast, and in response, transfers a request for wireless network slice **106** to control circuitry **104** over communication circuitry **105**. The wireless broadcast may indicate that wireless network slice **106** delivers a special communication service that drives wireless communication device **101** to request wireless network slice **106**.

[0030] Control circuitry **104** determines that the use of the wireless network slice **106** by wireless communication device **101** would be unauthorized. In some examples, no wireless communication devices are authorized to use wireless network slice **106**. In response to the slice request and the lack of authorization, control circuitry **104** directs communication circuitry **105** to serve wireless communication device **101** using wireless network slice **106**. Communication circuitry **105** directs wireless network slice **106** to serve wireless communication device **101**. Over communication circuitry **105**, control circuitry **104** directs wireless communication device **101** to use wireless network slice **106**.

[0031] Communication circuitry **105** serves wireless communication device **101** using wireless network slice **106** responsive to the direction of control circuitry **104**. Control circuitry **104** identifies a characteristic of wireless network slice **106** based on the service of wireless communication device **101** using wireless network slice **106**. Additional slice characteristics or patterns of slice characteristics may be determined by serving wireless communication devices **102-103** in a similar manner.

[0032] Advantageously, data communication system **100** efficiently identifies characteristics of unauthorized slice usage. Moreover, data communication system **100** effectively identifies patterns of the unauthorized slice usage across multiple wireless communication devices. Data communication system **100** protects network resources for legitimate users. Data communication system **100** maintains a good reputation and earns the trust of their customers.

[0033] FIG. 4 illustrates exemplary processing circuitry **400** to characterize use of wireless network slices. Processing circuitry **400** comprises an example of wireless communication devices **101-103**, control circuitry **104**, communication circuitry **105** and network slice **106**, although devices **101-103**, circuitry **104-105**, and slice **106** may differ. Processing circuitry **400** comprises machine-readable storage media **401-403** and microprocessors **407-409** that are communicatively coupled. Machine-readable storage media **401-403** store processing instructions **404-406** in a non-transitory manner. Microprocessors **407-409** comprise DSPs, CPUs, GPUs, ASICs, and/or some other data processing hardware. Machine-readable storage media **401-403** comprises RAM, flash circuitry, disk drives, and/or some other type of data storage apparatus. Microprocessors **407-409** retrieve processing instructions **404-406** from non-transitory machine-readable storage media **401-403**. Microprocessors **407-409** execute processing instructions **404-406** to characterize the unauthorized use of wireless network slices as described above for data communication system **100** and as described below for wireless communication network **500**. The amount of storage media, microprocessors, processing instructions that are shown in FIG. 4 may vary in other examples.

[0034] FIG. 5 illustrates exemplary wireless communication network **500** to characterize the unauthorized use of wireless network slice **510**. Wireless communication network **500** comprises an example of data communication system **100** and processing circuitry **400**, although system **100** and circuitry **400** may differ. Wireless communication system **500** comprises User Equipment (UE) **501-503**, Fifth Generation New Radio (5G NR) Access Node (AN) **504**, Wireless Fidelity (WIFI) AN **505**, Satellite (SAT) AN **506**, Satellite Ground Station (SAT GND) **507**, and Network Function Virtualization Infrastructure (NFVI) **508**. NFVI **508** comprises wireless network slice **510**, Interworking Function (IWF) **511**, Access and Mobility Management Function (AMF) **512**, Unified Data Management and Unified Data Repository (UDM/UDR) **513**, Network Slice Selection Function (NSSF) **514**, Application Server (AS) **515**, and Network Exposure Function (NEF) **516**. Wireless network slice **510** comprises Session Management Function (SMF) **517** and User Plane Function (UPF) **518**.

[0035] There are various scenarios when the use of wireless network slice **510** would be unauthorized. In some scenarios, the purpose of wireless network slice **510** is to characterize unauthorized use, so slice **510** does not have any authorized UEs, and any request by UEs **501-503** to use wireless network slice **510** would be unauthorized. In other scenarios, UEs **501-502** are subscribed to wireless network slice **510**, but UE **503** is not subscribed to slice **510**, so requests to use slice **510** by UE **503** would be unauthorized. In yet other scenarios, UEs **501-503** may be subscribed to wireless network slice **510**, but for some reason wireless communication network **510** instructs the UEs **501-503** not to use slice **510**, so requests to use wireless network slice **510** by UEs **501-503** after receiving such an instruction would be unauthorized. For example, wireless network slice **510** may not be available in the geographic area occupied by UEs **501-503**. Wireless network slice **510** may not be available due to maintenance, network outage, or some other reason, and requests to use slice **510** after such an instruction would be unauthorized. In yet another example, wireless communication network **500** may instruct UEs **501-503** not to request wireless network slice **510** when they are requesting or using another wireless network slice. Thus, the simultaneous use of both wireless network slices is not allowed, so a request by UEs **501-503** to use slice **510** when they are requesting or using the other slice would be unauthorized.

[0036] 5G NR AN **504** wirelessly broadcasts System Information Blocks (SIBs) that describe wireless network slice **510** by slice type, slice differentiator, and possibly other information. AMF **512** may also transfer slice type, slice differentiator, and other slice information to UEs **501-503** over Non-Access Stratum (NAS) messages that traverse one of the following paths: 1) 5G NR **504**, 2) IWF **511** and WIF AN **505**, and 3, IWF **511**, SAT GND **507**, and SAT AN **506**.

[0037] In operation, UE **501** transfers a request to AMF **512** over 5G NR AN **505** to use wireless network slice **510** in response to the SIBs and/or NAS messages. AMF **512** interacts with UDM/UDR **513** to determine if the request for slice **510** by UE **501** is authorized. In this example, UE **501** is not authorized because it is not subscribed to slice **510**. Based on rules for slice **510** in UDM/UDR **513**, UDM/UDR **513** directs AMF **512** to establish a data session for UE **501** to use slice **510** and collect slice usage data. Alternatively, AMF **512** may interact with NSSF **514** based on UDM/UDR data for UE **501** to determine that UE **501** is not authorized and to implement a data session to collect session data to characterize the unauthorized slice usage.

[0038] AMF **512** interacts with UDM/UDR **513**, SMF **517**, and typically other network functions to develop UE context for the data session like network addressing and quality-of-service levels. AMF **512** transfers the UE context to SMF **517** which forwards some UE context to UPF **518**. AMF **512** transfers the UE context to 5G NR AN **504**. AMF **512** also transfers some UE context to UE **501** over 5G NR AN **504**. AMF **512** may interact with UE **501** over NAS messaging that traverses 5G NR AN **505** to collect session information like the device model, operating system version, and user application that are related to the use of slice **510**.

[0039] UE **501** and UPF **518** exchange user data over 5G NR AN **504** per the UE context. UPF may exchange the user data with external data systems like web sites. UPF **518** analyzes the user data for session information like traffic types, destination network addresses, uniform resource indicators, and the like. AMF **512** collects session information from 5G NR AN **504** like geographic locations of UE **501**, UE mobility, radio status, and the like. AMF **512** also collects the session information from the UE context and UDM/UDR data for UE **501**. UE **501**, 5G NR **504**, UDM/UDR **513**, SMF **517**, and UPF **518** transfer their session information to AMF **512**. AMF **512** transfers the session information to UDM/UDR **513**.

[0040] After collecting the session information related to the use of wireless network slice **510** by UE **501**, AMF **512** signals 5G NR AN **504**, SMF **517**, and possibly UE **501** to terminate the data session. SMF **517** directs UPF **518** to terminate the data session. 5G NR AN **504** and UPF **518** stop the exchange of the user data between UE **501** and wireless network slice **510**.

[0041] AS **515** retrieves the session information for UE **501** from UDM/UDR **513**. AS **515** processes the session data to determine characteristics for the use of slice **510** like UE location, UE

operating system, UE user application, UE radio status, quality-of-service, network addresses, uniform resource indicators, traffic types, and the like. Subsequently, AS 515 aggregates the characteristics with those developed for UEs 502-503 to detect patterns of slice usage.

[0042] In another operation, slice 510 is not available in the tracking area occupied by UE 502, so AMF 512 instructs UE 502 not to request slice 510. Against this instruction, UE 502 still transfers a request to AMF 512 over WIFI AN 505 and IWF 511 to use wireless network slice 510—possibly in response to SIBs and/or NAS messages. AMF 512 interacts with UDM/UDR 513 to determine that the request for slice 510 by UE 502 is not authorized. Based on rules for slice 510 in UDM/UDR 513, UDM/UDR 513 directs AMF 512 to establish a data session for UE 502 to use slice 510 and collect slice usage data. Alternatively, AMF 512 may interact with NSSF 514 based on UDM/UDR data for UE 502 to determine that UE 502 is not authorized and to implement a data session to collect session data to characterize the unauthorized slice usage.

[0043] AMF 512 interacts with UDM/UDR 513, SMF 517, and typically other network functions to develop UE context for the data session like network addressing and quality-of-service levels. AMF 512 transfers the UE context to SMF 517 which forwards some UE context to UPF 518. AMF 512 transfers the UE context to IWF 511. AMF 512 also transfers some UE context to UE 502 over IWF 511 and WIFI AN 505. AMF 512 may interact with UE 501 over NAS messaging that traverses IWF 511 and WIFI AN 505 to collect session information like the device model, operating system version, and user application that are related to the use of slice 510.

[0044] UE 502 and UPF 518 exchange user data over WIFI AN 505 and IWF 511 per the UE context. UPF 518 may exchange the user data with external data systems like web sites. UPF 518 analyzes the user data for session information like traffic types, destination network addresses, uniform resource indicators, and the like. AMF 512 collects session information from the UE context and UDM/UDR data for UE 502. UE 502, UDM/UDR 513, SMF 517, and UPF 518 transfer their session information to AMF 512. AMF 512 transfers the session information to UDM/UDR 513.

[0045] After collecting the session information for the use of wireless network slice 510 by UE 502, AMF 512 signals IWF 511, SMF 517, and possibly UE 502 to terminate the data session. SMF 517 directs UPF 518 to terminate the data session. IWF 511 and UPF 518 stop the exchange of the user data between UE 502 and wireless network slice 510.

[0046] AS 515 retrieves the session information for UE 502 from UDM/UDR 513. AS 515 processes the session data to determine characteristics for the use of slice 510 by UE 502 like UE location, UE operating system, UE user application, UE quality-of-service, network addresses, uniform resource indicators, traffic types, and the like. Subsequently, AS 515 aggregates the characteristics with those developed for UEs 501 and 503 to detect patterns of slice usage.

[0047] In another operation, AMF 512 instructs UE 503 not to request slice 510 when using a different satellite-based slice. UE 503 is using the satellite based slice and against the instruction, UE 503 still transfers a request to AMF 512 over SAT AN 506, SAT GND 507, and IWF 511 to use wireless network slice 510—possibly in response to SIBs and/or NAS messages. AMF 512 interacts with UDM/UDR 513 to determine that the request for slice 510 by UE 503 is not authorized. Based on rules for slice 510 in UDM/UDR 513, UDM/UDR 513 directs AMF 512 to establish a data session for UE 503 to use slice 510 and collect slice usage data. Alternatively, AMF 512 may interact with NSSF 514 based on UDM/UDR data for UE 503 to determine that UE 503 is not authorized and to implement a data session to collect session data to characterize the unauthorized slice usage.

[0048] AMF 512 interacts with UDM/UDR 513, SMF 517, and typically other network functions to develop UE context for the data session like network addressing and quality-of-service levels. AMF 512 transfers the UE context to SMF 517 which forwards some UE context to UPF 518. AMF 512 transfers the UE context to IWF 511. AMF 512 also transfers some UE context to UE 503 over IWF 511, SAT GND 507, and SAT AN 506. AMF 512 may interact with UE 503 over

NAS messaging that traverses IWF 511, SAT GND 507, and SAT AN 506 to collect session information like the device model, operating system version, and user application that are related to the use of slice 510.

[0049] UE 501 and UPF 518 exchange user data over SAT AN 506, SAT GND 507, and IWF 511 per the UE context. UPF 518 may exchange the user data with external data systems like web sites. UPF 518 analyzes the user data for session information like traffic types, destination network addresses, uniform resource indicators, and the like. AMF 512 collects session information from the UE context and UDM/UDR data for UE 503. UE 502, UDM/UDR 513, SMF 517, and UPF 518 transfer their session information to AMF 512. AMF 512 transfers the session information to UDM/UDR 513.

[0050] After collecting the session information for the use of wireless network slice 510 by UE 503, AMF 512 signals IWF 511, SMF 517, and possibly UE 501 to terminate the data session. SMF 517 directs UPF 518 to terminate the data session. IWF 511 and UPF 518 stop the exchange of the user data between UE 503 and wireless network slice 510.

[0051] AS 515 retrieves the session information for UE 503 from UDM/UDR 513. AS 515 processes the session data to determine characteristics for the use of slice 510 by UE 503 like UE location, UE operating system, UE user application, UE quality-of-service, network addresses, uniform resource indicators, traffic types, and the like. Subsequently, AS 515 aggregates the characteristics with those developed for UEs 501-502 to detect patterns of slice usage. For example, AS 515 may detect a common user application and operating system version in UEs 501-503 when they use slice 510 in an unauthorized manner. In another example, AS 515 may detect a common geographic location and device type for UEs 501-503 when they use slice 510 in an unauthorized manner.

[0052] In another operation, NEF 516 stores rules for slice 510 and UEs 501-503. UE 503 transfers a request to AMF 512 over SAT AN 506, SAT GND 507, and IWF 511 to use wireless network slice 510—possibly in response to SIBs and/or NAS messages. AMF 512 interacts with NEF 516 to determine that the request for slice 510 by UE 503 is not authorized. Based on rules for slice 510 and UE 503, NEF 516 directs AMF 512 to establish a data session for UE 503 to use slice 510 and collect slice usage data. AMF 512 interacts with UDM/UDR 513, SMF 517, and typically other network functions to develop UE context for the data session like network addressing and quality-of-service levels. AMF 512 transfers the UE context to SMF 517 which forwards some UE context to UPF 518. AMF 512 transfers the UE context to IWF 511. AMF 512 also transfers some UE context to UE 503 over IWF 511, SAT GND 507, and SAT AN 506. AMF 512 may interact with UE 503 over NAS messaging that traverses IWF 511, SAT GND 507, and SAT AN 506 to collect session information like the device model, operating system version, and user application that are related to the use of slice 510.

[0053] UE 501 and UPF 518 exchange user data over SAT AN 506, SAT GND 507, and IWF 511 per the UE context. UPF 518 may exchange the user data with external data systems like web sites. UPF 518 analyzes the user data for session information like traffic types, destination network addresses, uniform resource indicators, and the like. AMF 512 collects session information from the UE context and UDM/UDR data for UE 503. UE 502, UDM/UDR 513, SMF 517, and UPF 518 transfer their session information to AMF 512. AMF 512 transfers the session information to UDM/UDR 513.

[0054] After collecting the session information for the use of wireless network slice 510 by UE 503, AMF 512 signals IWF 511, SMF 517, and possibly UE 501 to terminate the data session. SMF 517 directs UPF 518 to terminate the data session. IWF 511 and UPF 518 stop the exchange of the user data between UE 503 and wireless network slice 510.

[0055] NEF 516 retrieves the session information for UE 503 from UDM/UDR 513. NEF 516 processes the session data to determine characteristics for the use of slice 510 by UE 503 like UE location, UE operating system, UE user application, UE quality-of-service, network addresses,

uniform resource indicators, traffic types, and the like. Subsequently, NEF **516** aggregates the characteristics with those developed for UEs **501-502** to detect patterns of slice usage. For example, NEF **516** may detect a common user application and geographic location for UEs **501-503** when they use slice **510** in an unauthorized manner.

[0056] FIG. **6** illustrates exemplary UE **501** in wireless communication network **500** that characterizes the unauthorized use of wireless network slice **510**. UE **501** comprises an example of wireless communication devices **101-103** and UEs **502-503**, although devices **101-103** and UEs **502-503** may differ. UE **501** comprises Fifth Generation New Radio (5G NR) radio circuitry **601**, Wireless Fidelity (WIFI) radio circuitry **602**, satellite radio circuitry **603**, and processing circuitry **604**. Radio circuitry **601-603** comprises antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSPs, memories, and transceivers (XCVRs) that are coupled over bus circuitry. Processing circuitry **604** comprises one or more CPUs, one or more memories, and one or more transceivers that are coupled over bus circuitry. The one or more memories in processing circuitry **604** store software like an Operating System (OS), 5G NR Application (5G NR), 3GPP Application (3GPP), WIFI Application (WIFI), Satellite Application (SAT), and Internet Protocol application (IP). The antennas in radio circuitry **601-603** exchange wireless signals with ANs **504-506**. Transceivers in radio circuitry **601-603** are coupled to transceivers in processing circuitry **604**. In processing circuitry **604**, the one or more CPUs retrieve the software from the one or more memories and execute the software to direct the operation of UE **501** as described herein.

[0057] FIG. **7** illustrates exemplary Fifth Generation New Radio (5G NR) Access Node (AN) **504** in wireless communication network **500** that characterizes the unauthorized use of wireless network slice **510**. 5G NR AN **504** comprises an example of communication circuitry **105** and processing circuitry **400**, although circuitry **105** and circuitry **400** may differ. 5G NR AN **504** comprises 5G NR Radio Unit (RU) **701**, Distributed Unit (DU) **702**, and Centralized Unit (CU) **703**. 5G NR RU **701** comprises antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSP, memory, radio applications, and transceivers that are coupled over bus circuitry. DU **702** comprises memory, CPU, user interfaces and components, and transceivers that are coupled over bus circuitry. The memory in DU **702** stores operating system and 5G NR network applications for Physical Layer (PHY), Media Access Control (MAC), and Radio Link Control (RLC). CU **703** comprises memory, CPU, and transceivers that are coupled over bus circuitry. The memory in CU **703** stores an operating system and 5G NR network applications for Packet Data Convergence Protocol (PDCP), Service Data Adaptation Protocol (SDAP), and Radio Resource Control (RRC). The antennas in 5G NR RU **701** are wirelessly coupled to UEs **501-503** over 5G NR links. Transceivers in 5G NR RU **701** are coupled to transceivers in DU **702**. Transceivers in DU **702** are coupled to transceivers in CU **703**. Transceivers in CU **703** are coupled to transceivers in NFVI **508**. The DSP and CPU in RU **701**, DU **702**, and CU **703** execute the radio applications, operating systems, and network applications to exchange data and signaling between UEs **501-503** and NFVI **508** as described herein.

[0058] FIG. **8** illustrates exemplary Wireless Fidelity (WIFI) AN **505** in wireless communication network **500** that characterizes the unauthorized use of wireless network slice **510**. WIFI AN **505** comprises an example of communication circuitry **105** and processing circuitry **400**, although circuitry **105** and circuitry **400** may differ. WIFI AN **505** comprises WIFI radio **801** and processing circuitry **802**. Radio **801** comprises antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSPs, memories, and transceivers that are coupled over bus circuitry. Processing circuitry **802** comprises one or more CPUs, one or more memories, and one or more transceivers that are coupled over bus circuitry. The one or more memories in processing circuitry **802** store software like an Operating System (OS), WIFI application (WIFI), and IP application (IP). The antennas in WIFI radio **801** exchange WIFI signals with UEs **501-503**. Transceivers in radio **801** are coupled to transceivers in processing circuitry **802**. Transceivers in processing circuitry **802** are coupled to transceivers in NFVI **508**. In processing circuitry **802**, the one or more CPUs retrieve

the software from the one or more memories and execute the software to exchange data and signaling between UEs **501-503** and NFVI **508** as described herein.

[0059] FIG. **9** illustrates exemplary SAT AN **506** and SAT GND **507** in wireless communication network **500** that characterizes the unauthorized use of wireless network slice **510**. SAT AN **506** and SAT GND **507** comprise examples of communication circuitry **105** and processing circuitry **400**, although circuitry **105** and circuitry **400** may differ. SAT AN **506** comprises UE radio **901**, GND radio **902** and processing circuitry **903**. SAT GND **507** comprises satellite radio **904** and processing circuitry **905**. Radios **901-902** and **904** comprise antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSPs, memories, and transceivers that are coupled over bus circuitry. Processing circuitry **903** and **905** comprise one or more CPUs, one or more memories, and one or more transceivers that are coupled over bus circuitry. The one or more memories in processing circuitry **903** and **905** store software like an Operating System (OS), Satellite Application (SAT), and IP Application (IP). The antennas in UE radio **901** exchange satellite signals with UEs **501-503**. Transceivers in UE radio **901** are coupled to transceivers in processing circuitry **903**. Transceivers in processing circuitry **903** are coupled to transceivers in GND radio **902**. The antennas in GND radio **902** exchange satellite signals with antennas in satellite radio **904**, and the antennas in satellite radio **904** exchange the satellite signals with GND radio **902**. Transceivers in satellite radio **904** are coupled to transceivers in processing circuitry **905**. Transceivers in processing circuitry **905** are coupled to transceivers in NFVI **508**. In processing circuitry **903** and **905**, the one or more CPUs retrieve the software from the one or more memories and execute the software to exchange data and signaling between UEs **501-503** and NFVI **508** as described herein.

[0060] FIG. **10** illustrates exemplary Network Function Virtualization Infrastructure (NFVI) **508** in wireless communication network **500** that characterizes the unauthorized use of wireless network slice **510**. NFVI **508** comprises an example of control circuitry **104**, communication circuitry **105**, and processing circuitry **400**, although circuitry **104-105** and circuitry **400** may differ. NFVI **508** comprises hardware **1001**, hardware drivers **1002**, operating systems **1003**, virtual layer **1004**, and network functions **1005**. Hardware **1001** comprises Network Interface Cards (NICS), CPUS, RAM, Flash/Disk Drives (DRIVES), and Data Switches (DSWS). Hardware drivers **1002** comprise software that is resident in the NICS, CPUS, RAM, DRIVES, and DSWS. Operating systems **1003** comprise kernels, modules, applications, and containers. Virtual layer **1004** comprises virtual Operating Systems (vOS), vNICS, vCPUS, vRAM, vDRIVES, and vSWS. Network Functions **1005** comprises Slice SW **1010**, IWF SW **1011**, AMF SW **1012**, UDM/UDR SW **1013**, NSSF SW **1014**, AS SW **1015**, and NEF SW **1016**. Slice SW **1010** comprises SMF SW **1017** and UPF SW **1018**. The NICS in hardware **1001** are coupled to ANs **504-505**, SAT GND **507**, and external systems. Hardware **1001** executes hardware drivers **1002**, operating systems **1003**, virtual layer **1004**, and network functions **1005** to form and operate IWF **511**, AMF **512**, UDM/UDR **513**, NSSF **514**, AS **515**, NEF **516**, SMF **517**, and UPF **518** as described herein. NFVI **508** comprises one or more microprocessors and one or more non-transitory machine-readable storage media that store processing instructions that direct NFVI **508** to exchange data and signaling between ANs **504-505**, SAT GND **507**, and external systems as described herein. NFVI **508** may be located at a single site or be distributed across multiple geographic locations.

[0061] FIG. **11** illustrates an exemplary operation of wireless communication network **500** to characterize the unauthorized use of wireless network slice **510**. The operation may differ in other examples. 5G NR AN **504** wirelessly broadcasts information related to slice **510** like slice type and differentiator. UE **501** receives the broadcast slice information, and in response, transfers a request (RQ) for slice **510** to AMF **512** over 5G NR AN **504**. AMF **512** retrieves UE information for UE **501** from UDM/UDR **513**. The UE information indicates that AMF **512** should monitor UE **501** when it attempts to use slice **510**. AMF **512** develops UE context for UE **501** to use slice **510** like network addressing and quality-of-service levels. AMF **512** transfers the UE context with

instructions to monitor and report on UE **501** to SMF **517** and 5G NR AN **504**. SMF **517** transfers the UE context with instructions to monitor and report on UE **501** to UPF **518**. AMF **512** transfers the UE context to UE **501** over 5G NR AN **504**. In response to the UE context, UE **501** exchanges user data with UPF **518** over 5G NR AN **504**.

[0062] UE **501**, 5G NR AN **504**, UDM/UDR **513**, SMF **517**, and UPF **518** generate session information related to the use of slice **510** by UE **501** and transfer the session information to AMF **512**. The session information indicates the user application and operating system version being used by UE **501** to use slice **510**. The session information indicates the network addresses and uniform resource indicator being used by UE **501** to use slice **510**. The session information indicates the quality-of-service levels and geographic locations being used by UE **501** to use slice **510**. AMF **512** processes the session information to generate slice usage data for UE **501** and slice **510**. AMF transfers the slice usage data to UDM/UDR **513**. AMF **512** signals 5G NR AN **504** and SMF **517** to terminate the data session between UE **501** and slice **510**. SMF **517** signals UPF **518** to terminate the data session between UE **501** and slice **510**. 5G NR AN **504** and UPF **518** stop the exchange of user data between UE **501** and slice **510**.

[0063] Although not shown, UEs **502-503** are handled in the same manner and their slice usage information is also transferred to UDM/UDR **513** by AMF **512**. UDM/UDR **513** aggregates this slice usage data for UEs **501-503** and transfers the aggregated slice usage data to AS **515**. AS **515** processes the slice usage data for slice **510** to determine characteristics and patterns related to the slice usage. For example, AS **515** may detect a common operating system version that uses slice **510** in an unauthorized manner.

[0064] FIG. **12** illustrates an exemplary operation of wireless communication network **500** to characterize the unauthorized use of wireless network slice **510**. The operation may differ in other examples. 5G NR AN **504** wirelessly broadcasts information related to slice **510** like slice type and differentiator. UE **501** receives the broadcast slice information, and in response, transfers a request for slice **510** to AMF **512** over 5G NR AN **504**. AMF **512** retrieves UE information for UE **501** from UMD/UDR **513**. The UE information indicates that the request from UE **501** to use slice **510** is rejected. AMF **512** queries NSSF **514** for slice selection assistance for UE **501** and slice **510**. NSSF **514** responds to AMF **512** that UE **501** is not authorized to use slice **510** but that AMF **512** should monitor UE **501** when it uses slice **510**.

[0065] AMF **512** develops UE context for UE **501** to use slice **510** like network addressing and quality-of-service levels. AMF **512** transfers the UE context with instructions to monitor and report on UE **501** to SMF **517** and 5G NR AN **504**. SMF **517** transfers the UE context with instructions to monitor and report on UE **501** to UPF **518**. AMF **512** transfers the UE context to UE **501** over 5G NR AN **504**. In response to the UE context, UE **501** exchanges user data with UPF **518** over 5G NR AN **504**.

[0066] UE **501**, 5G NR AN **504**, UDM/UDR **513**, SMF **517**, and UPF **518** generate session information related to the use of slice **510** by UE **501** and transfer the session information to AMF **512**. The session information indicates the user application and operating system version being used by UE **501** to use slice **510**. The session information indicates the network addresses and uniform resource indicator being used by UE **501** to use slice **510**. The session information indicates the quality-of-service levels and geographic locations being used by UE **501** to use slice **510**. AMF **512** processes the session information to generate slice usage data for UE **501** and slice **510**. AMF transfers the slice usage data to UDM/UDR **513**. AMF **512** signals 5G NR AN **504**, and SMF **517** to terminate the data session between UE **501** and slice **510**. SMF **517** signals UPF **518** to terminate the data session between UE **501** and slice **510**. 5G NR AN **504** and UPF **518** stop the exchange of user data between UE **501** and slice **510**.

[0067] Although not shown, UEs **502-503** are handled in the same manner and their slice usage information is also transferred to UDM/UDR **513** by AMF **512**. UDM/UDR **513** aggregates this slice usage data for multiple UEs and transfers the aggregated slice usage data to AS **515**. AS **515**

processes the slice usage data for slice **510** to determine characteristics and patterns related to the slice usage. For example, AS **515** may detect a common user application that uses slice **510** in an unauthorized manner.

[0068] FIG. **13** illustrates an exemplary operation of wireless communication network **500** to characterize the unauthorized use of wireless network slice **510**. The operation may differ in other examples. 5G NR AN **504** wirelessly broadcasts information related to slice **510** like slice type and differentiator. UE **501** receives the broadcast slice information, and in response, transfers a request for slice **510** to AMF **512** over 5G NR AN **504**. AMF **512** retrieves UE information for UE **501** from UDM/UDR **513**. The UE information indicates that the request from UE **501** to use slice **510** is rejected. AMF **512** queries NEF **514** for slice control assistance for UE **501** and slice **510**. NEF **514** applies user permissions, network policies, and resource availability to determine that UE **501** should not be authorized to use slice **510** but that AMF **512** should monitor UE **501** when it uses slice **510**.

[0069] AMF **512** develops UE context for UE **501** to use slice **510** like network addressing and quality-of-service levels. AMF **512** transfers the UE context with instructions to monitor and report on UE **501** to SMF **517** and 5G NR AN **504**. SMF **517** transfers the UE context with instructions to monitor and report on UE **501** to UPF **518**. AMF **512** transfers the UE context to UE **501** over 5G NR AN **504**. In response to the UE context, UE **501** exchanges user data with UPF **518** over 5G NR AN **504**.

[0070] UE **501**, 5G NR AN **504**, UDM/UDR **513**, SMF **517**, and UPF **518** generate session information related to the use of slice **510** by UE **501** and transfer the session information to AMF **512**. The session information indicates the user application and operating system version being used by UE **501** to use slice **510**. The session information indicates the network addresses and uniform resource indicator being used by UE **501** to use slice **510**. The session information indicates the quality-of-service levels and geographic locations being used by UE **501** to use slice **510**. AMF **512** processes the session information to generate slice usage data for UE **501** and slice **510**. AMF transfers the slice usage data to NEF **516**. AMF **512** signals 5G NR AN **504**, and SMF **517** to terminate the data session between UE **501** and slice **510**. SMF **517** signals UPF **518** to terminate the data session between UE **501** and slice **510**. 5G NR AN **504** and UPF **518** stop the exchange of user data between UE **501** and slice **510**.

[0071] Although not shown, UEs **502-503** are handled in the same manner and their slice usage information is also transferred to UDM/UDR **513** by AMF **512**. UDM/UDR **513** aggregates this slice usage data for multiple UEs and transfers the aggregated slice usage data to NEF **516**. NEF **516** processes the slice usage data for slice **510** to determine characteristics and patterns related to the slice usage. For example, NEF **156** may detect a common user application that uses slice **510** in an unauthorized manner.

[0072] Advantageously, wireless communication network **500** efficiently identifies characteristics of unauthorized slice usage. Moreover, wireless communication network **500** effectively identifies patterns of the unauthorized slice usage across multiple UEs. Wireless communication network **500** protects network resources for legitimate users. Wireless communication network **500** maintains a good reputation and earns the trust of their customers.

[0073] The wireless communication system circuitry described above comprises computer hardware and software that form special-purpose data communication circuitry to characterize unauthorized use of wireless network slices. The computer hardware comprises processing circuitry like CPUs, DSPs, GPUs, transceivers, bus circuitry, and memory. To form these computer hardware structures, semiconductors like silicon or germanium are positively and negatively doped to form transistors. The doping comprises ions like boron or phosphorus that are embedded within the semiconductor material. The transistors and other electronic structures like capacitors and resistors are arranged and metallurgically connected within the semiconductor to form devices like logic circuitry and storage registers. The logic circuitry and storage registers are arranged to form larger

structures like control units, logic units, and Random-Access Memory (RAM). In turn, the control units, logic units, and RAM are metallically connected to form CPUs, DSPs, GPUs, transceivers, bus circuitry, and memory.

[0074] In the computer hardware, the control units drive data between the RAM and the logic units, and the logic units operate on the data. The control units also drive interactions with external memory like flash drives, disk drives, and the like. The computer hardware executes machine-level software to control and move data by driving machine-level inputs like voltages and currents to the control units, logic units, and RAM. The machine-level software is typically compiled from higher-level software programs. The higher-level software programs comprise operating systems, utilities, user applications, and the like. Both the higher-level software programs and their compiled machine-level software are stored in memory and retrieved for compilation and execution. On power-up, the computer hardware automatically executes physically-embedded machine-level software that drives the compilation and execution of the other computer software components which then assert control. Due to this automated execution, the presence of the higher-level software in memory physically changes the structure of the computer hardware machines into special-purpose data communication circuitry system to characterize unauthorized use of wireless network slices.

[0075] The above description and associated figures teach the best mode of the invention. The following claims specify the scope of the invention. Note that some aspects of the best mode may not fall within the scope of the invention as specified by the claims. Those skilled in the art will appreciate that the features described above can be combined in various ways to form multiple variations of the invention. Thus, the invention is not limited to the specific embodiments described above, but only by the following claims and their equivalents.

Claims

1. A method, the method comprising: receiving requests from wireless communication devices to use a wireless network slice; determining that the use of the wireless network slice by the wireless communication devices would be unauthorized, and in response, serving the wireless communication devices using the wireless network slice; and identifying a characteristic of the wireless network slice responsive to serving the wireless communication devices using the wireless network slice.
2. The method of claim 1 further comprising: wirelessly broadcasting slice information related to the wireless network slice; and wherein the wireless communication devices request the wireless network slice responsive to the wireless broadcast.
3. The method of claim 1 further comprising: wirelessly broadcasting slice information related to a high-throughput service provided by the wireless network slice; and wherein the wireless communication devices request the wireless network slice responsive to the slice information.
4. The method of claim 1 further comprising: wirelessly broadcasting slice information related to a low-latency service provided by the wireless network slice; and wherein the wireless communication devices request the wireless network slice responsive to the slice information.
5. The method of claim 1 further comprising: wirelessly broadcasting slice information related to a high-definition service provided by the wireless network slice; and wherein the wireless communication devices request the wireless network slice responsive to the slice information.
6. The method of claim 1 further comprising: wirelessly broadcasting slice information related to a gaming service provided by the wireless network slice; and wherein the wireless communication devices request the wireless network slice responsive to the slice information.
7. The method of claim 1 further comprising: wirelessly broadcasting slice information related to a telepresence service provided by the wireless network slice; and wherein the wireless communication devices request the wireless network slice responsive to the slice information.

- 8.** The method of claim 1 wherein identifying the characteristic of the wireless network slice comprises identifying one or more types of the wireless communication devices.
 - 9.** The method of claim 1 wherein identifying the characteristic of the wireless network slice comprises identifying one or more user applications in the wireless communication devices.
 - 10.** The method of claim 1 wherein identifying the characteristic of the wireless network slice comprises identifying one or more operating systems in the wireless communication devices.
 - 11.** The method of claim 1 wherein identifying the characteristic of the wireless network slice comprises identifying one or more destination network addresses used by the wireless communication devices.
 - 12.** The method of claim 1 wherein identifying the characteristic of the wireless network slice comprises identifying one or more uniform resource indicators used by the wireless communication devices.
 - 13.** The method of claim 1 wherein identifying the characteristic of the wireless network slice comprises identifying one or more geographic locations of the wireless communication devices.
 - 14.** The method of claim 1 wherein no other wireless communication devices are authorized to use the wireless network slice.
 - 15.** A method, the method comprising: wirelessly broadcasting slice information related to a wireless network slice and receiving requests from wireless communication devices to use the wireless network slice, wherein the wireless communication devices request the wireless network slice responsive to the wireless broadcast; determining that the use of the wireless network slice by the wireless communication devices would be unauthorized, and in response, serving the wireless communication devices using the wireless network slice; and identifying a characteristic of the wireless network slice responsive to serving the wireless communication devices using the wireless network slice, wherein the characteristic of the wireless network slice comprises at least one of: types of the wireless communication devices, user applications in the wireless communication devices, operating systems in the wireless communication devices, destination network addresses used by the wireless communication devices, uniform resource indicators used by the wireless communication devices, and geographic locations of the wireless communication devices.
 - 16.** A data communication system comprising: control circuitry to determine that the use of a wireless network slice by wireless communication devices would be unauthorized, and in response, to direct communication circuitry to serve the wireless communication devices using the wireless network slice; communication circuitry to serve the wireless communication devices using the wireless network slice responsive to the control circuitry; and the control circuitry to identify a characteristic of the wireless network slice based on the service of the wireless communication devices using the wireless network slice.
 - 17.** The data communication system of claim 16 further comprising: wirelessly broadcasting slice information related to the wireless network slice; and wherein the wireless communication devices request the wireless network slice responsive to the wireless broadcast.
 - 18.** The data communication system of claim 16 wherein identifying the characteristic of the wireless network slice comprises identifying one or more types of the wireless communication devices.
 - 19.** The data communication system of claim 16 wherein identifying the characteristic of the wireless network slice comprises identifying one or more user applications in the wireless communication devices.
 - 20.** The data communication system of claim 16 wherein no other wireless communication devices are authorized to use the wireless network slice.
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